

Goutham Darapogu

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SUMMARY:

I'm a Software Engineer with experience in designing and implementing backend systems, data engineering pipelines, and cloud-based software systems. Expertise in building scalable APIs, distributed computing, and real-time data infrastructure using programming languages such as Python and Java and cloud services including AWS and Kubernetes.

EDUCATION:

Masters in Computer Science

[Aug 2024 - May 2026]

Texas A&M Corpus Christi GPA:3.92/4.0

Btech in Computer Science Engineering

[Aug 2018 - May 2022]

Indian institute of information technology Sri city, chitoor

WORK EXPERIENCE:

Graduate Research assistant, Texas A&M Corpus Christi

[Sept 2025 - May 2026]

- Extraction and automated processing of 10k+ LinkedIn and SEC documents using Python and Selenium for extensive financial analysis
- Development of event studies and sentiment analyses pipelines for evaluating investor behavior and layoff trends within the workforce
- Enhancement of data processing pipelines using Pandas and Regular Expressions leading to increase in efficiency by 35%
- Conversion of structured raw data into modelled data enhancing research effectiveness by 25%
- GitHub copilot used to facilitate programming and debug tests performed on code

Student Worker, Texas A&M Corpus Christi

[Aug 2024 - Jun 2025]

- Created Python-based back-end automation scripts for processing and manipulation of large datasets, which were used in analytics and reporting systems
- Created data processing pipelines that worked in an Extract-Transform-Load pattern, to automate processing, validation, and formatting of data sets, thus increasing scalability of workflows
- Automated backend processes of report creation and generation through data manipulation of data sets with the help of back-end scripting tools
- Optimized data processing operations through the use of tools like Pandas and Regex for increased efficiency and reliability of the workflow
- Worked on projects associated with scalable workflows, back-end automation, handling of structured data, debugging, and optimization

Software Engineer MBS Software Bangalore

[Sept 2022 – May 2024]

- Developed and maintained backend components for data-driven web applications using Python, APIs, CSV, and JSON-based data integration workflows
- Optimized queries and multithreading workflows, thereby increasing application performance and efficiency by 25%
- Designed analytics and report dashboards in Python with help from libraries such as Pandas, Matplotlib, and Seaborn
- Created workflows for logging and monitoring using AWS CloudWatch for identifying problems in production environments and saving time in debugging by 20%
- Enabled data processing and automation pipelines by creating Python workflows for back-end processing

Software Engineer, Cambridge mobile telematics

[Jan 2022 – Aug 2022]

- Enhanced web application roles for improved user accessibility, authentication, and access control through Django frameworks
- Designed and deployed an ML-based anomaly detection system using Scikit-learn that has minimized false positives by 15% via data analysis and validation processes
- Deployed containerized and scalable backend applications on Docker and Amazon Web Services services like EC2 and S3 with 50K+ users supported per day
- Automated CI/CD processes utilizing GitHub Actions and minimized deployment time by 35%
- Established logging, monitoring, and debugging processes that have enabled the proactive identification of

problems and improved application reliability by 25%

Machine Learning Volunteer Developer

[Jan 2021 – Dec 2021]

- Created an algorithm to detect fake news using Python programming language along with Scikit-learn and NLP techniques to categorize the fake and genuine news articles on the basis of textual datasets
- Created a complete pipeline for data cleaning and preprocessing involving vectorization via TF-IDF, text tokenization, stopwords elimination, and text normalization for large textual news datasets
- Created several classification algorithms like logistic regression, random forest, and support vector machine (SVM), which were trained and tested to obtain 87% accuracy with lower false positives
- Implemented the data preprocessing steps automatically utilizing Pandas and NumPy libraries, making experimental work easier and lowering down preprocessing time by 30%
- Enhanced model performance via hyperparameters tuning, validation testing, and feature selection techniques

PROJECTS:

StreamFlow – Real-Time Data Engineering Platform

- Created and deployed an end-to-end real-time data engineering solution using Kafka, Airflow, PostgreSQL, and Streamlit for efficient stream processing, workflow orchestration, and analysis
- Created pipelines to extract, transform, and load cryptocurrency market data from live data streams, incorporating workflow scheduling and real-time updates
- Created visual dashboards for analyzing real-time market trends and providing insights through back-end data stream processing workflows

Distributed Task Queue & Notification System

- Developed a distributed system that processes tasks using FastAPI, Redis, RabbitMQ, and PostgreSQL, designed to deal with async job execution and notifications
- Implemented REST APIs, scheduled tasks, retries, and worker functionality that allowed building a scalable and robust backend
- Created a pipeline for task execution tracking and system performance monitoring using Docker and AWS

InsightForge AI — Multi-Agent RAG Platform for Enterprise Knowledge Automation

- Created and implemented multi-agent RAG framework based on LangChain, FastAPI, OpenAI APIs, and Pinecone for intelligence-based enterprise document search and contextual questioning
- Developed scalable pipelines for document ingestions, embeddings, vector search, and semantic retrieval processes to manage large unstructured knowledge datasets
- Implemented containerized AI-based systems on AWS cloud services using Docker and Kubernetes, incorporating asynchronous inferencing pipelines along with hallucination and response evaluation systems

Multi-Cloud Resource Optimizer

- Created a cloud-based resource optimization tool using Python, Docker, and Kubernetes to handle scalable workload management on the AWS and GCP platforms
- Applied live monitoring and auto-scaling functionality to enhance computing resource utilization efficiency and achieve a 20% reduction in infrastructure expenditure
- Implemented Kubernetes technology to deploy containers and enable scalable and efficient cloud applications

TECHNICAL SKILLS:

Languages: Python, Java, SQL, JavaScript, TypeScript, C++, C#

Backend & Software Engineering: Spring Boot, Django, FastAPI, REST APIs, API Integration, Backend Development, Microservices, Distributed Systems, OOP, System Design, Multithreading, Concurrent Programming, Asynchronous Processing

Frontend & Full-Stack: MERN Stack, React.js, Node.js, [Express.js](#)

Cloud & Infrastructure: AWS (S3, EC2, Lambda, CloudWatch), GCP, Docker, Kubernetes, Terraform, CI/CD, Container Orchestration, Monitoring & Logging

Data Engineering: ETL/ELT, Data Pipelines, Apache Airflow, Kafka, Apache Spark, Workflow Orchestration, Batch Processing, Real-Time Streaming, Data Processing

GenAI & Machine Learning: LangChain, OpenAI APIs, RAG Pipelines, Vector Databases (Pinecone/FAISS/ChromaDB), Scikit-learn, PyTorch, Pandas, NumPy, NLP, GitHub Copilot

Databases & Tools: PostgreSQL, MongoDB, MySQL, Git, Linux